📘 AZ-104 – LAB 04 TASK 3 & TASK 4 – DETAILED NOTES (SECURITY + DNS) 🔐 TASK 3: CONFIGURE COMMUNICATION BETWEEN ASG AND NSG 🎯 Objective of Task 3

The goal of Task 3 is to secure network traffic inside the Virtual Network by:

Controlling which resources can communicate

Restricting which ports are allowed

Blocking unnecessary internet access

Making security rules scalable and readable

In real Azure environments, network security is enforced at the subnet and NIC level, not inside applications.

🧠 Why Azure Uses NSG and ASG

Azure separates security enforcement and workload grouping:

Component Purpose NSG Enforces traffic rules ASG Groups workloads logically

This separation makes security scalable and maintainable.

🔹 Network Security Group (NSG) – Deep Explanation

An NSG is a stateful firewall that filters traffic using rules.

Key properties of NSG rules:

Evaluated by priority

Lower number = higher priority

First matching rule is applied

Evaluation stops after a match

Default NSG behavior:

Inbound: Deny all

Outbound: Allow all

This means:

If you don’t explicitly allow traffic, it is blocked.

🔹 Application Security Group (ASG) – Deep Explanation

An ASG groups VMs by application role, not IP address.

Why ASG is important:

IP addresses can change

VM scale sets add/remove instances

ASGs stay consistent even when infrastructure changes

Example:

asg-web → Web servers asg-db → Database servers

Security rules reference roles, not IPs.

🧱 What We Implemented in Task 3 1️⃣ Created ASG: asg-web

Purpose:

Represents web-tier workloads

Used as source in NSG rules

At this stage:

No VM is attached

Still valid and expected

2️⃣ Created NSG: myNSGSecure

Purpose:

Acts as subnet-level firewall

Controls inbound and outbound traffic

3️⃣ Associated NSG with SharedServicesSubnet

This is a critical step.

Why?

NSG only works after association

Subnet-level NSG affects all resources inside it

🔐 Inbound Rule – Allow Web Traffic from ASG Rule logic: Allow traffic FROM asg-web ON ports 80 and 443 USING TCP

Why ports 80 and 443?

80 → HTTP

443 → HTTPS

Most web applications use these ports

Why Priority = 100?

Ensures rule is evaluated before default deny

Leaves space for future rules

Follows industry best practice

🚫 Outbound Rule – Deny Internet Access Rule logic: Deny ALL outbound traffic TO Internet ON all ports

Why deny outbound internet?

Prevents malware communication

Prevents data exfiltration

Forces traffic through controlled paths (firewall, proxy)

Why destination ports = 0–65535?

Means all ports

Equivalent to *

Why Priority = 4096?

Higher than inbound allow rules

Lower than default allow (65001)

Overrides default outbound internet access

🔍 Final Security Result (Task 3)

✔ Only allowed traffic enters subnet ✔ Internet access is blocked ✔ Security is role-based ✔ Rules are scalable and readable

🌐 TASK 4: CONFIGURE PUBLIC AND PRIVATE AZURE DNS 🎯 Objective of Task 4

The goal of Task 4 is to enable name resolution, so resources can be accessed using:

hostname.domain.com

instead of:

IP addresses

DNS is mandatory for real-world cloud environments.

🔹 Why DNS is Critical in Azure

Without DNS:

Applications break when IP changes

Configurations become fragile

Scaling becomes difficult

With DNS:

Applications use names

Infrastructure changes are hidden

Environments scale smoothly

🌍 Public DNS Zone – Deep Explanation What Public DNS does:

Resolves names from the internet

Used for:

Websites

APIs

Public services

Example: www.contosolab.com → Public IP

🧱 Public DNS Record (A Record)

An A record maps:

hostname → IPv4 address

Example:

www.contosolab.com → 10.1.1.4

(Used as demo value in lab)

🔐 Private DNS Zone – Deep Explanation

Private DNS:

Works only inside VNets

Is not accessible from the internet

Used for internal services

Example:

sensorvm.private.contosolab.com → 10.1.1.4

🔗 Virtual Network Link – Why It's Required

Private DNS zones do NOTHING until linked.

The link:

Tells Azure which VNet can use this DNS

Enables name resolution inside that network

Without linking: ❌ DNS records exist ❌ But cannot be resolved

🧾 Private DNS Recordsets – Why We Added Them

A recordset defines:

“When someone asks for this name, return this IP.”

Without recordsets:

DNS zone exists

But resolves nothing

Recordsets make DNS functional.

🔍 Final DNS Result (Task 4)

✔ Public names resolve externally ✔ Private names resolve internally ✔ VNets use DNS without exposing services ✔ IP changes won’t break applications

🎯 INTERVIEW-READY MASTER ANSWERS

Q: Why use ASG instead of IP addresses? A: ASGs group workloads logically and remain stable as infrastructure scales.

Q: Why is NSG stateful? A: Return traffic is automatically allowed once a connection is established.

Q: Why deny outbound internet traffic? A: To reduce attack surface and prevent data exfiltration.

Q: Why use private DNS instead of public DNS internally? A: To keep internal services hidden from the internet.

Q: Why link private DNS to VNet? A: Without linking, name resolution does not work.

☑ FINAL KEY TAKEAWAY (IMPORTANT)

Task 3 secures how traffic flows Task 4 controls how resources are found

Together, they form the core of Azure network architecture.